

README.md

Multiple Linear Regression – Multi-Channel Marketing Analysis

Project Overview

This project demonstrates how to build and evaluate a **Multiple Linear Regression** model using Python to analyze the impact of marketing expenditures across multiple advertising channels on product sales. The analysis focuses on determining how investments in **TV**, **Radio**, and **Social Media** advertising contribute to sales performance while addressing multicollinearity and validating the assumptions of linear regression.

The project is designed as a comprehensive educational resource for students studying **Data Science**, **Business Analytics**, **Machine Learning**, and **Applied Statistics**. It emphasizes statistical reasoning, model interpretation, and business decision-making using real-world analytical techniques.

Project Objectives

The objectives of this project are to:

- Build a Multiple Linear Regression model using Python and Statsmodels.
 - Create and analyze a realistic marketing dataset.
 - Perform exploratory data analysis (EDA).
 - Detect multicollinearity using correlation analysis and Variance Inflation Factor (VIF).
 - Evaluate model performance using Adjusted R-squared and statistical significance tests.
 - Validate regression assumptions using residual diagnostic plots.
 - Interpret regression coefficients in a business context.
 - Develop evidence-based recommendations for marketing budget allocation.
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Business Problem

A company's marketing department invests advertising budgets across multiple channels:

- TV Advertising
- Radio Advertising
- Social Media Advertising

Management wants to understand:

- Which advertising channels have the greatest influence on sales?

- Are any marketing channels highly correlated with each other?
- Which variables should remain in the predictive model?
- How should future marketing budgets be allocated to maximize sales?

This project answers these questions using Multiple Linear Regression.

Dataset Description

A realistic synthetic marketing dataset is generated within the notebook and saved as a CSV file.

Variables

Variable	Description
TV	TV advertising expenditure (in thousands of dollars)
Radio	Radio advertising expenditure (in thousands of dollars)
Social_Media	Social media advertising expenditure (in thousands of dollars)
Sales	Product sales (in thousands of units)

Example:

TV	Radio	Social_Media	Sales
120.5	35.2	48.1	22.8
180.7	42.3	65.4	30.5
95.4	18.7	30.9	17.3

Project Workflow

The notebook follows a structured data science workflow:

1. Import Required Libraries

- pandas
- numpy
- matplotlib

- seaborn
 - scipy
 - statsmodels
 - sklearn
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2. Generate a Realistic Marketing Dataset

- Simulate advertising budgets
 - Introduce realistic relationships among variables
 - Generate sales values with random variation
 - Save the dataset as a CSV file
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3. Exploratory Data Analysis (EDA)

- Load the dataset
 - Display sample records
 - Inspect data types
 - Check for missing values
 - Generate descriptive statistics
 - Visualize variable distributions
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4. Correlation Analysis

- Compute Pearson correlation coefficients
 - Visualize relationships using a heatmap
 - Identify potential multicollinearity
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5. Multicollinearity Detection

Calculate the Variance Inflation Factor (VIF) for each predictor.

Interpretation:

- $VIF = 1 \rightarrow$ No multicollinearity
- $VIF < 5 \rightarrow$ Acceptable
- $VIF > 5 \rightarrow$ Moderate multicollinearity

- $VIF > 10 \rightarrow$ Serious multicollinearity
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6. Build the Multiple Linear Regression Model

Fit an Ordinary Least Squares (OLS) regression model using Statsmodels.

Model equation:

$$\text{Sales} = \beta_0 + \beta_1(\text{TV}) + \beta_2(\text{Radio}) + \beta_3(\text{Social Media}) + \epsilon$$

7. Model Evaluation

Evaluate the model using:

- R-squared
 - Adjusted R-squared
 - F-statistic
 - p-values
 - Confidence intervals
 - Standard errors
-

8. Regression Diagnostics

Validate model assumptions using:

- Residual vs Fitted Plot
- Histogram of Residuals
- Normal Q-Q Plot
- Scale-Location Plot

These diagnostics help assess:

- Linearity
- Normality
- Homoscedasticity
- Constant variance
- Outliers

9. Coefficient Interpretation

Interpret each regression coefficient while holding the remaining variables constant.

Example:

Holding Radio and Social Media spending constant, every additional **\$1,000** invested in TV advertising is associated with an average increase of **X** thousand units in sales.

10. Business Recommendations

Based on statistical evidence, recommend:

- Which advertising channels deserve increased investment
 - Which channels provide limited return
 - Whether any predictors should be excluded due to statistical insignificance
 - An optimized marketing budget strategy
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Expected Learning Outcomes

By completing this project, students will learn how to:

- Perform multiple linear regression analysis.
 - Generate and analyze realistic datasets.
 - Detect multicollinearity using VIF.
 - Interpret regression coefficients correctly.
 - Evaluate statistical significance using hypothesis testing.
 - Understand Adjusted R-squared.
 - Validate regression assumptions.
 - Translate statistical findings into business recommendations.
 - Build reproducible data science workflows.
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Project Structure

Multiple_Linear_Regression_Marketing_Analysis/

```
|
|├── data/
|│   └── marketing_data.csv
|├── notebooks/
|│   └── multiple_regression_analysis.ipynb
|├── images/
|│   ├── correlation_heatmap.png
|│   ├── residual_plot.png
|│   ├── histogram_residuals.png
|│   ├── qq_plot.png
|│   └── scale_location_plot.png
|├── README.md
|├── requirements.txt
|└── LICENSE
```

Installation

Clone the repository:

```
git clone
https://github.com/samuel1599361/Multiple_Linear_Regression_Marketing_Analysis.git
```

```
cd Multiple_Linear_Regression_Marketing_Analysis
```

Install the required packages:

```
pip install pandas, numpy, matplotlib, seaborn, scipy, statsmodels, scikit-learn, jupyter
```

Alternatively, install from the requirements file:

```
pip install -r requirements.txt
```

Running the Project

Launch Jupyter Notebook:

```
jupyter notebook
```

Open:

notebooks/multiple_regression_analysis.ipynb

Run all notebook cells sequentially to reproduce the analysis.

Required Python Libraries

- pandas
 - numpy
 - matplotlib
 - seaborn
 - scipy
 - statsmodels
 - scikit-learn
 - jupyter
-

Example Results

Typical outputs include:

- Correlation Matrix
 - Correlation Heatmap
 - VIF Table
 - OLS Regression Summary
 - Adjusted R-squared
 - Predictor Significance Tests
 - Residual Diagnostic Plots
 - Business Insights
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Key Skills Demonstrated

- Data Cleaning
- Exploratory Data Analysis (EDA)
- Feature Analysis
- Correlation Analysis
- Multiple Linear Regression
- Statistical Modeling
- Hypothesis Testing
- Model Evaluation

- Multicollinearity Detection
- Residual Diagnostics
- Business Analytics
- Data Visualization
- Statistical Interpretation

Applications

The techniques demonstrated in this project can be applied in:

- Marketing Analytics
- Sales Forecasting
- Business Intelligence
- Financial Modeling
- Customer Analytics
- Advertising Optimization
- Retail Analytics
- Revenue Prediction
- Operations Research
- Strategic Decision-Making

Future Enhancements

Potential extensions include:

- Feature engineering
- Polynomial regression
- Interaction effects between advertising channels
- Cross-validation
- Regularized regression (Ridge and Lasso)
- Automated feature selection
- Model deployment using Flask or FastAPI
- Interactive dashboards with Plotly or Streamlit

Conclusion

This project provides a complete, end-to-end implementation of Multiple Linear Regression for marketing analytics. It demonstrates how statistical modeling can be used to quantify the impact of multiple advertising channels on sales while addressing multicollinearity, validating model assumptions, and translating analytical findings into actionable business

recommendations. The workflow reflects industry best practices and serves as a strong foundation for more advanced predictive modeling and business analytics projects.

License

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Data Science | Machine Learning | Business Analytics | Artificial Intelligence

Feel free to contribute by submitting issues, feature requests, or pull requests. Feedback and suggestions are always welcome.